HGK
Aug 19 58

identify

Invert Matrix by Cramer's rule
Remultiply ? as check ?

A

A^{-1}

Difference Matrix between A^{-1} w/o noise A^{-1} noise

no noise

A_n^{-1}

Δ matrix

$A_n^{-1} - A^{-1}$

or percentage of A ?
or no. of places correct ?

a measure of precision
what is best ?
one . no of digits correct.

A^*

obtained by mpy back.

Δ^*

$A^* - A$

average Δ

4 × 4 Matrix Inversion by Cramer's Rule

$$[a]^{-1} = \frac{[A_{ji}]}{|a|}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$$

Matrix in row order

Transpose:

$$\begin{vmatrix} a_{11} & a_{21} & a_{31} & a_{41} \\ a_{12} & a_{22} & a_{32} & a_{42} \\ a_{13} & a_{23} & a_{33} & a_{43} \\ a_{14} & a_{24} & a_{34} & a_{44} \end{vmatrix}$$

$A^{\circ}_{ji} = A'_{ij}$ example take a_{23}

$$A'_{23} = \begin{vmatrix} a_{11} & a_{31} & a_{41} \\ a_{12} & a_{32} & a_{42} \\ a_{14} & a_{34} & a_{44} \end{vmatrix} = a_{11}\begin{vmatrix} a_{32} & a_{42} \\ a_{34} & a_{44} \end{vmatrix} - a_{12}\begin{vmatrix} a_{31} & a_{41} \\ a_{34} & a_{44} \end{vmatrix} + a_{14}\begin{vmatrix} a_{31} & a_{41} \\ a_{32} & a_{42} \end{vmatrix}$$

$$A'_{24} = \begin{vmatrix} a_{11} & a_{31} & a_{41} \\ a_{12} & a_{32} & a_{42} \\ a_{13} & a_{33} & a_{43} \end{vmatrix} = a_{11}\begin{vmatrix} a_{32} & a_{42} \\ a_{33} & a_{43} \end{vmatrix} - a_{12}\begin{vmatrix} a_{31} & a_{41} \\ a_{33} & a_{43} \end{vmatrix} + a_{13}\begin{vmatrix} a_{31} & a_{41} \\ a_{32} & a_{42} \end{vmatrix}$$

A_{23}

```
STZ   S2
STZ   S1

LDQ   a44        a44      a42
FMP   a32        a31      a31
FAD   S1
STO   S1

LDQ   a42        a41      a41
FMP   -a34       a34      a32
FAD   S1
STO   S1

LDQ   S1
FMP   a11        -a12     +a14
FAD   S2
STO   S2
```

STO A_{23}

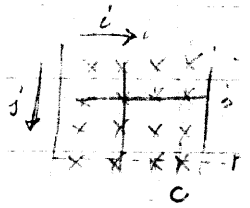
set I=m, J=m
calc R, C
set i=m, j=m
calc r_i c_j

$m \to r$
Test $r = \underset{\text{on } I}{i}$ if so set, $r-1 \to r$
$m \to c$
Test $c = \underset{\text{on } J}{j}$ if so set $c-1 \to c$

$r-1 \to r_1$
Test $r_1 = \underset{\text{on } I}{i}$ if so set $r_1 - 1 \to r$
$c-1 \to c_1$
Test $c_1 = \underset{\text{on } J}{j}$ if so set $c_1 - 1 \to c$

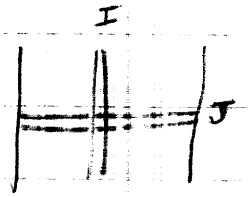
calc 2 term + add to sum

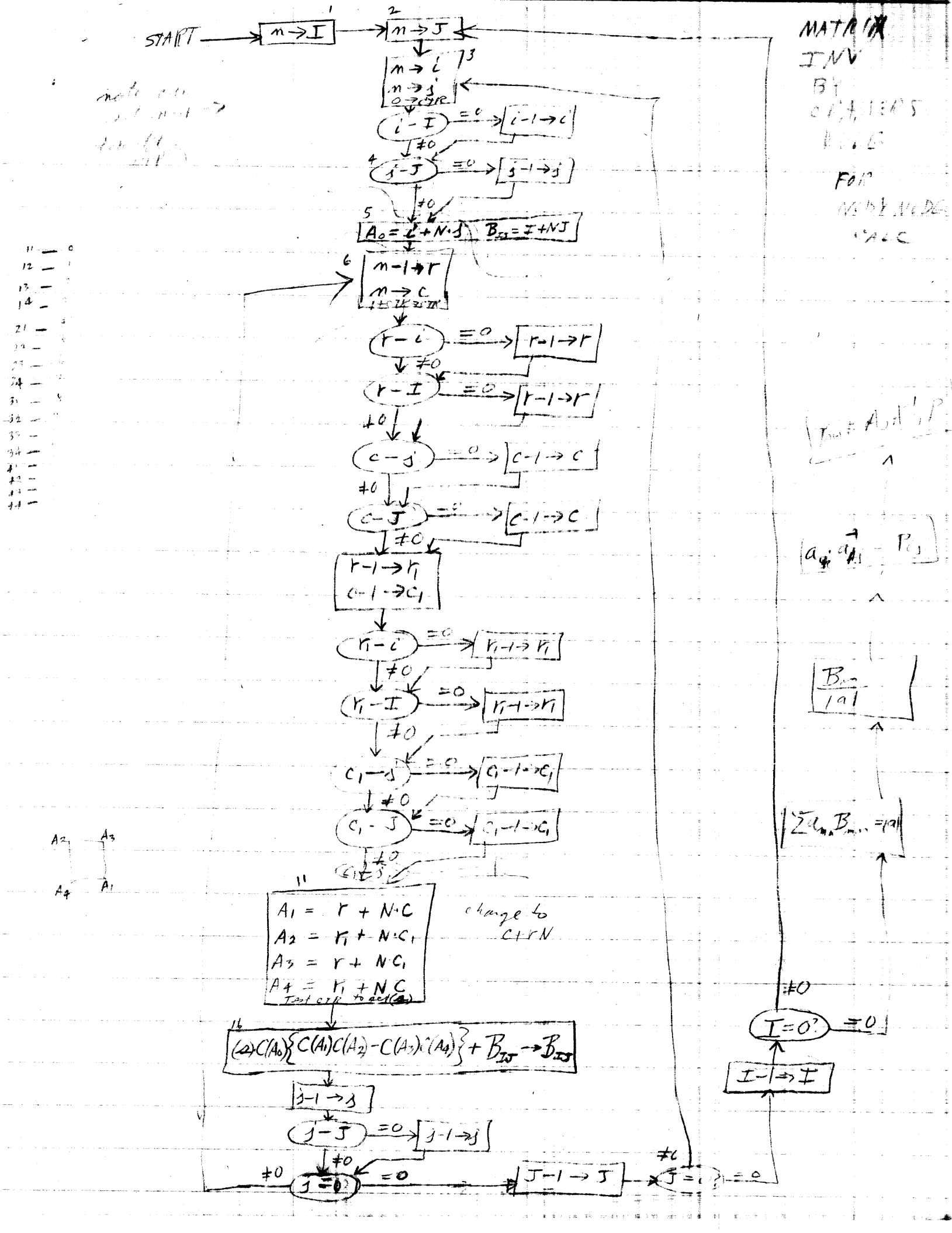
adv. $j-1 \to j$
Test if $\neq 0$ → reset $j=m$ $i-1 \to i$
Test if $i=0$ → end

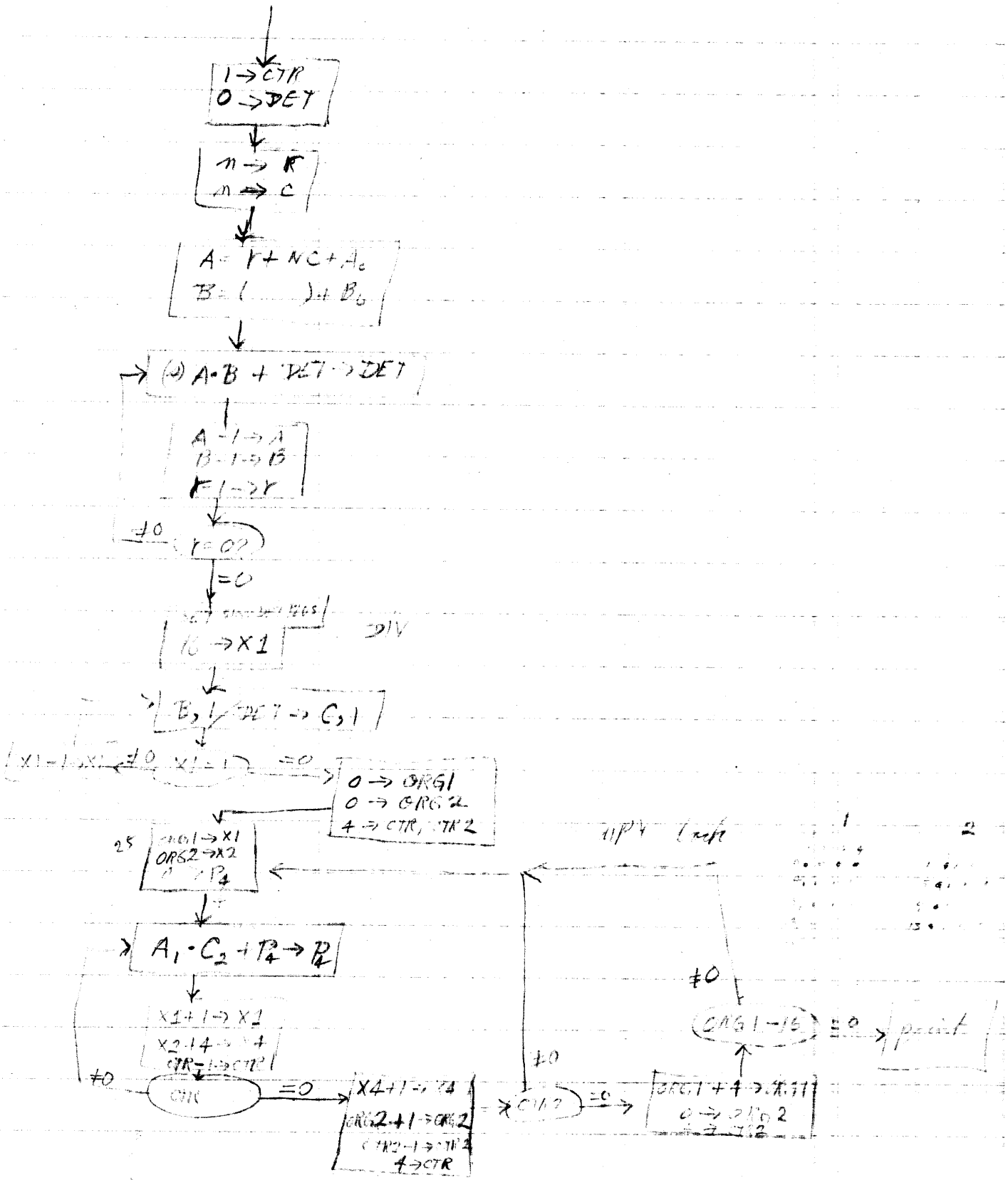


$m \to r$
$m \to c$
$r-i$
$r-I$

$$\begin{pmatrix} + & - & + \\ + & + & - \\ + & + & + \end{pmatrix}$$




START
1
n → I
2
n → J
3
n → i
n → j
0 → cycle
i - I
=0
i-1 → i
≠0
4
j - J
=0
j-1 → j
≠0
5
A₀ = i + N·j
B_ij = I + NJ
6
n-1 → r
n → c
r - i
=0
r-1 → r
≠0
r - I
=0
r-1 → r
≠0
c - j
=0
c-1 → c
≠0
c - J
=0
c-1 → c
≠0
r-1 → r₁
c-1 → c₁
r₁ - i
=0
r₁-1 → r₁
≠0
r₁ - I
=0
r₁-1 → r₁
≠0
c₁ - j
=0
c₁-1 → c₁
≠0
c₁ - J
=0
c₁-1 → c₁
≠0
11
A₁ = r + N·C
A₂ = r₁ + N·C₁
A₃ = r + N·C₁
A₄ = r₁ + N·C
change to C₁ r N
16
(-1)^{i+j} C(A₀) { C(A₁)C(A₂) - C(A₃)C(A₄) } + B_ij → B_ij
j-1 → j
j - J
=0
j-1 → j
≠0
j = 0?
≠0
=0
J-1 → J
J = 1?
≠0
=0
I-1 → I
I = 0?
≠0
=0
MATRIX
INV
FOR
Σ a_ik B_kj = 1 if i = j, 0 if i ≠ j
A₂ A₃
A₄ A₁


1 → CTR
0 → DET
n → R
n → C
A = 1 + NC + A_c
B = ( )+ B_0
(2) A·B + DET → DET
A · 1 → A
B · 1 → B
R / 2 → R
≠0
R = 0?
=0
10 → X1
DIV
B_1 / DET → C_1
X1 = 1?
≠0
=0
0 → ORG1
0 → ORG2
4 → CTR, CTR2
25
ORG1 → X1
ORG2 → X2
A_1 · C_2 + P_4 → P_4
X1 + 1 → X1
X2 · 4 → X4
CTR − 1 → CTR
CTR
≠0
=0
X4 + 1 → X4
ORG2 + 1 → ORG2
CTR2 − 1 → CTR2
4 → CTR
CTR2
=0
ORG1 + 4 → X111
0 → 2/0 2
2 → CTR2
≠0
ORG1 − 16
=0
print
1
2

4 x 4 address scheme

IJ	i j	r c	r_1 c_1
44	33	22	11
	32	23	11
	31	23	12
43	34	22	11
	32	24	11
	31	24	12
42	34	23	11
	33	24	11
	31	24	13
41	34	23	12
	33	24	12
	32	24	13

IJ	i j	r c	r_1 c_1
34	43	22	11
	42	23	11
	41	23	12
33	44	22	11
	42	24	11
	41	24	12
32	44	23	11
	43	24	11
	41	24	13
31	44	23	12
	43	24	12
	42	24	13

IJ	i j	r c	r_1 c_1
24	43	32	11
	42	33	11
	41	33	12
23	44	32	11
	42	34	11
	41	34	12
22	44	33	11
	43	34	11
	41	34	13
21	44	33	12
	43	34	12
	42	34	13

IJ	i j	r c	r_1 c_1
14	43	32	21
	42	33	21
	41	33	22
13	44	32	21
	42	34	21
	41	34	22
12	44	33	21
	43	34	21
	41	34	23
11	44	33	22
	43	34	22
	42	34	23